Stefan Rogers-Coltman

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Neuroscientist with a PhD from the MRC Laboratory of Molecular Biology, bringing strong practical experience in end-to-end design and development. Skilled in combining scientific domain knowledge with hardware, electronics, and software engineering to deliver reliable, integrated systems. Experienced collaborating with engineering specialists across disciplines, with a track record of rapidly engaging with unfamiliar technical domains.

Education

PhD, Systems Neuroscience *2021–2025*
University of Cambridge, Cambridge, UK
Supervisor: Dr. Marco Tripodi, MRC Laboratory of Molecular Biology.

MSc, Translational Neuroscience *2020–2021*
Imperial College London, London, UK

BSc (Hons), Biomedical Sciences *2017–2020*
University College London, London, UK
1st Class Honours. Neuroscience specialisation.

Research Experience

Postdoctoral Researcher *2025–present*
MRC Laboratory of Molecular Biology, Cambridge, UK
Extending the behavioural platform developed during my PhD, working across the scientist-engineer interface and supervising a computing student.

- Led development of a fully wireless successor to the behavioural rig system detailed below, taking it from concept to manufactured hardware in weeks: designed custom PCBs and wrote MCU firmware (STM32, BLE)
- Designed and built a mouse-wearable wireless motion sensor (9-DOF IMU with magnetometer)

Doctoral Researcher *2021–2025*
MRC Laboratory of Molecular Biology, Cambridge, UK
Initiated and led a cross-functional research programme, developing a scalable experimental platform integrating hardware, software, and neuroscience methods, now adopted across multiple projects within the lab. Collaborated closely with researchers and mechanical/electrical workshops to design, build, and deploy robust experimental systems.

Experimental Systems & Engineering

- Designed and built automated behavioural rigs for studying spatial decision-making in freely moving mice
- Developed wireless load-cell platforms (ESP32, Bluetooth) for automated trial initiation
- Implemented high-speed video acquisition systems in C++ using Spinnaker API, enabling synchronised multi-camera recording
- Engineered general-purpose data acquisition architecture for synchronising behavioural, video, and electrophysiology signals, with end-to-end pipelines for storage and analysis
- Built integrated optogenetics control systems combining IMU-based head tracking, automated laser stimulation, and real-time data visualisation

Systems Neuroscience & Circuit Mechanisms

- Developed a novel behavioural paradigm dissociating sensory input from motor output in a naturalistic setting
- Manipulated superior colliculus Pitx2+ neurons using chemogenetics (DREADDs), demonstrating modulation of baseline spatial priors across movements
- Performed chronic electrophysiological recordings (optetrodes, Cambridge NeuroTech, Neuropixels) in freely moving mice during behaviour
- Identified a subpopulation of neurons encoding spatial confidence, selectively active during correct visual decisions but not during incorrect or non-visual trials
- Integrated neural and behavioural datasets to link circuit-level activity with decision-making processes

Selected Additional Experience

Teaching Assistant *2026*
Cambridge Open Lab: Open Science in Practice Summer School,
Taught engineering techniques for neuroscience to PhD students, assisting practical workshops in electronics design (Arduino) and CAD (FreeCAD).

Head of Maintenance (Bosun) *2026*
Cambridge University Yacht Club,
Responsible for maintenance and operational readiness of club yacht. Diagnosing and resolving mechanical and electrical issues to ensure safe and continuous operation.

MSc & BSc Research Projects *2019–21*
Imperial College London / UCL, London, UK
Developed experimental systems for generating senescent cell cultures (MSc); investigated noradrenergic modulation of Golgi cell inhibition of cerebellar granule cells (BSc).

Founder *2020*
FPV Drone Build Venture,
Designed and built custom FPV drones; managed end-to-end assembly and client delivery.

Research Assistant *Summers 2018/19*
MRC LMB / UCL, Cambridge & London, UK
- Developed organoid culture techniques (inner ear and choroid plexus models)
- Performed imaging and analysis; contributed to protocol optimisation and automated sampling approaches

Skills

Programming & Software	Python (data analysis, automation), C++ (high-performance imaging), embedded firmware (STM32, BLE, Arduino)
Experimental Systems & Engineering	Behavioural rig design, PCB design, wireless sensor systems, real-time data acquisition, high-speed imaging (FLIR / Spinnaker), hardware–software integration, multi-system synchronisation
Neuroscience & *In Vivo* Methods	UK Home Office Personal Licence (PIL) holder; behavioural task design and animal training; mouse neurosurgery (viral injections, implants); chronic electrophysiology (optetrodes, Neuropixels)
Data Analysis	Statistical modelling, custom pipeline development, behavioural analysis, neural data integration

Publications

Manuscript in preparation — behavioural paradigm and superior colliculus manipulation study (Rogers-Coltman S, Welch D, Geyer L, Tripodi M.).

Awards & Funding

Laidlaw Undergraduate Research and Leadership Scholarship
Laidlaw Foundation *Funded research placements and leadership training*